

Shafiq Ur Rehman, Ph.D.

Assistant Professor (adjunk), Physics | University of Warsaw
Computational physicist and materials science engineer



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37	950	15	DFT MD GW+BSE 2D Materials Photoelectrochemistry
Articles	Citations	H-index	Core expertise

Profile

Computational condensed matter physicist specializing in first-principles calculations, GW+BSE methods, molecular dynamics, and multiscale simulations for renewable-energy materials, photoelectrochemical water splitting, 2D materials, heterostructures, and optoelectronic systems.

Employment

2025 - Present	Assistant Professor (adjunk), Physics University of Warsaw, Warsaw, Mazovia, Poland Academic employment in physics.
2022 - 2024	Postdoctoral Research Fellow Yangtze River Delta Institute, University of Electronic Science and Technology of China Research on optoelectronic and photocatalytic properties of 2D materials.
2021 - 2022	Research Associate Southern University of Science and Technology of China Optical excitation, doping, and defects in 2D composite materials.
2018 - 2020	Postdoctoral Research Fellow College of Optoelectronic Engineering, Shenzhen University Optoelectronic properties of 2D heterojunctions and Li-ion battery applications.

Education

2014 - 2018	Ph.D., Condensed Matter Physics University of Science and Technology of China, Hefei First-principles studies of 2D heterostructures for optoelectronic and renewable-energy applications.
2011 - 2014	M.Phil., Physics Hazara University, Mansehra, Pakistan Optical and electronic properties of semiconductor nanomaterials.
2009 - 2011	M.Sc., Physics (Gold Medalist) Hazara University, Mansehra, Pakistan Theoretical investigations of nanomaterials for spintronic applications.
2007 - 2009	B.Sc., Physics and Mathematics University of Peshawar, Pakistan

Technical Skills

VASP	Quantum ESPRESSO	LAMMPS
COMSOL	Python / Linux	Mathematica

Research Focus

Photocorrosion inhibition and photoelectrochemical water splitting	2D materials, TMD monolayers, and in-plane heterostructures
Electronic, optical, magnetic, and catalytic properties	Charge transport, defect dynamics, and excitonic effects
Li-ion battery anode materials and energy storage	Simulation-experiment synergy for materials design

Selected Publications

- **Unraveling the photocatalytic potential of transition metal sulfide and selenide monolayers for overall water splitting and photo-corrosion inhibition.** J. Mater. Chem. A 12, 6693-6702 (2024).
- **In-plane heterostructures of transition metal dichalcogenide monolayers with enhanced charge separation and effective overall water splitting.** Int. J. Hydrogen Energy 80, 280-288 (2024).
- **Computational insight of ZrS₂-graphene heterobilayer as an efficient anode material.** Applied Surface Science 551, 149304 (2021).
- **Enhanced electronic and optical properties of three TMD heterobilayers.** Phys. Chem. Chem. Phys. 20, 16604-16614 (2018).

Teaching

2021	Computational Materials Science Course-2 Southern University of Science and Technology, Shenzhen, China Computational materials science, DFT workflows, and simulation practice.
2020	Computational Materials Science Course-1 Shenzhen University, Shenzhen, China Supercomputer use, VASP inputs/outputs, DOS, band structure, Raman spectra, phonons, transition states, AIMD, quasiparticle and exciton effects.

Selected Talks

2026	Photocorrosion-Inhibited MoS₂-Based Heterostructures for Stable Photoanodes in Neutral pH Overall Water Splitting Seminar series, Modeling of Complex Systems, Warsaw, Poland
2021	Role of in-plane interface in controlling exciton recombination and enhanced photocatalytic properties UK-Canada Joint Symposium on Coordination Chemistry, St Andrews, UK
2019	Realization of noble heterobilayers with enhanced optoelectronic properties Hilton Garden Inn Houston Westbelt, Houston, USA

Achievements

- Chinese Government Scholarship Award, fully funded Ph.D. studies at USTC (2014-2018).
- Gold Medal Award in M.Sc. Physics, highest score among successful candidates (2011).
- Third position in Ph.D. topic presentation competition, Physical Academic Forum of USTC (2017).
- Best Reviewer Award from IOP Society, UK.
- Research Associate Professor Award, Zhejiang Province, China.